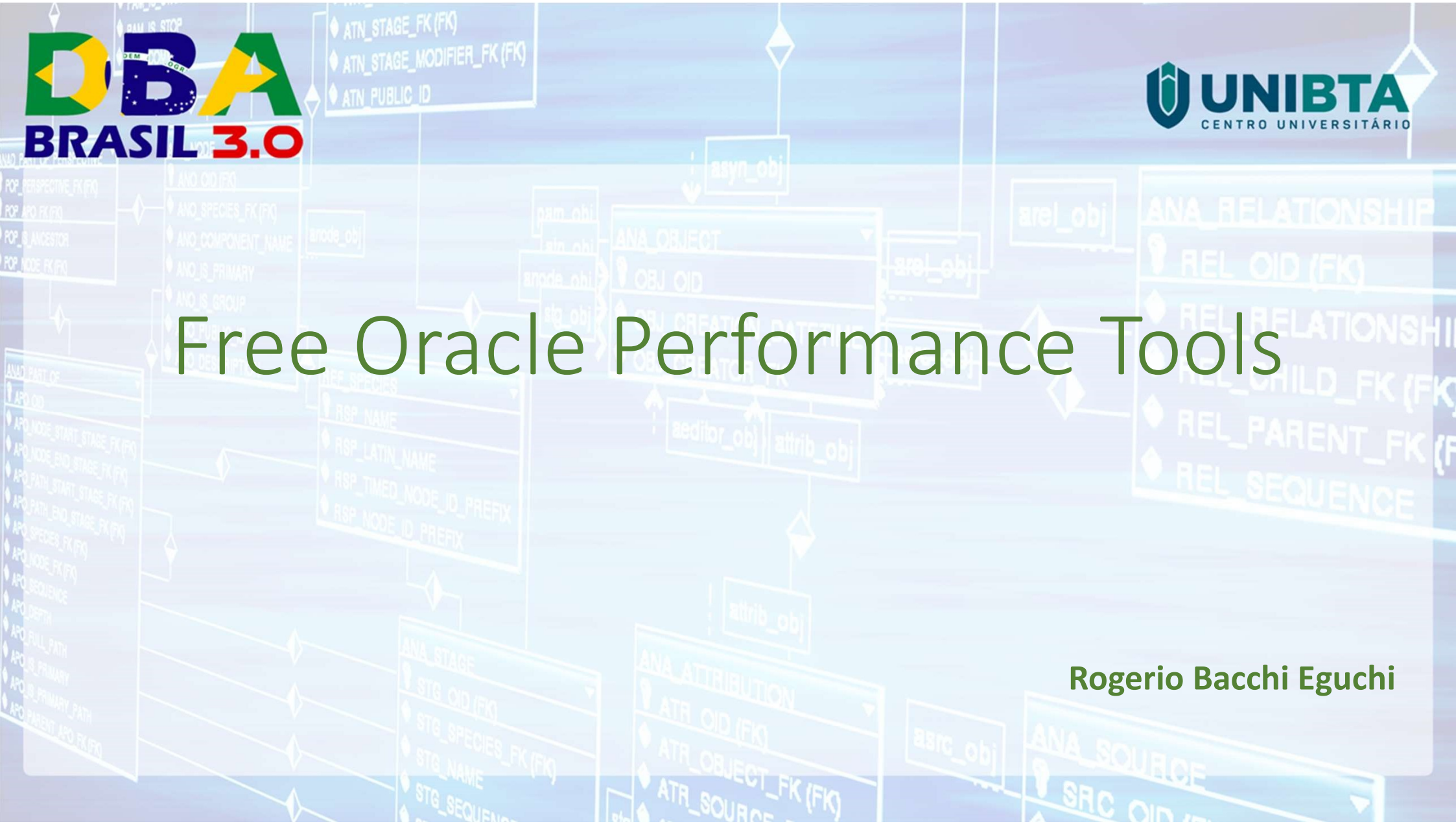



# Free Oracle Performance Tools

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NOSSOS PATROCINADORES
DELL EMC
TmaxSoft
Brasil
Timbira
A empresa brasileira de PostgreSQL
IS
InterSolution
The Right Choice
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solutions
soluções em tecnologia da informação
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Smooth Database Operation
ERPRIME
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DBmaster
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# Apresentação Pessoal

- Oracle DBA Sênior com extensa experiência em ambientes OLTP de missão crítica e que empregam as tecnologias da *Oracle Maximum Availability Architecture*
- 4 anos como DBA no Oracle Advanced Customer Support Services (2002-2005)
- 10 anos como DBA no UOL/UOLDiveo/PagSeguro (2005-2015)
- 2 anos como DBA/DMA na TOTVS (2016 > ...)
- Oracle 8i-12c Certified Professional. Oracle Exadata Certified Expert.
- RedHat 5 Certified Engineer
- Artigos publicados no OTN Brasil: 9 artigos
- Blogger: http://reguchi.wordpress.com
- Twitter: reguchi_br

# Agenda

1. Tuning – A history
2. Tuning – Objectives
3. Tuning – Pareto Principle
4. Real-time analysis
5. Historic Performance analysis
6. SQL tuning
7. SO tools

# Tuning - History

## Counters/Ratios

- Introduction of Wait Event instrumentation @Oracle 7

## Time-Based tuning

- YAPP Method (Yet Another Performance Profiling Method)
- Method R
- DB Time Tuning

# Tuning - History

| Oracle Version | Tool | Cost |
|---|---|---|
| 7 / 8 | utlbstat/utlestat | Free |
| 8i (8.1.6+) / 9i | Statspack | Free |
| 10g or > | Statspack<br><br>Diagnostic / Tuning Pack* | Free<br><br>US$ 60K/license** |

*Diagnostic and Tuning Pack are database options available only on Oracle Enterprise Edition

** US$60K = 1 Oracle EE + 1 Diag+Tuning Pack License

# Tuning - Objectives

1. Define performance and performance problems
2. Check the performance of the Oracle DB
3. Check the overall performance of the server and storage

# Tuning - Objectives

Why do I care about time ?

- Human time is critical to the enterprise
- Systems performance affects business goals
- “Time is money”
- Performance improvement means doing things faster

Performance is always and only about time

# Tuning - Objectives

Response Time = Work done (service time) + Waiting for work (wait time)

Each server process is typically in one of three states:

- Idle: Waiting for something to do (sleeping)
- Running code: Using the CPU or in a run queue
- Waiting: blocked
  - For some resource to become available
  - For a requested activity to complete

If service time and/or wait time is high, it will directly impact the total response time

# Tuning - Objectives

1. Reactive tuning
   - Firefighter
2. Proactive tuning
   - Plan, Do, Check, Act

# Pareto Principle (80/20 rule)

*"The minority of causes, inputs, or efforts usually produces a majority of the results"*

Top Foreground Events by Total Wait Time

SQL ordered by CPU Time

SQL ordered by User I/O Wait

SQL ordered by buffer gets

SQL ordered by physical reads

SQL ordered by Executions

# Real-time analysis – Snapper

## Snapper – Tanel Poder

- Realtime session monitor
- No installation needed
- Snaps v$session etc, reports the difference

## Arguments

1. Mode (ash, stat, custom)
2. Seconds to run
3. Number of runs
4. All or filter on v$session (ex: username = scott or sid=51)

# Real-time analysis – Snapper

```
SQL> @tp\snapper ash 5 1 all
Sampling SID all with interval 5 seconds, taking 1 snapshots...

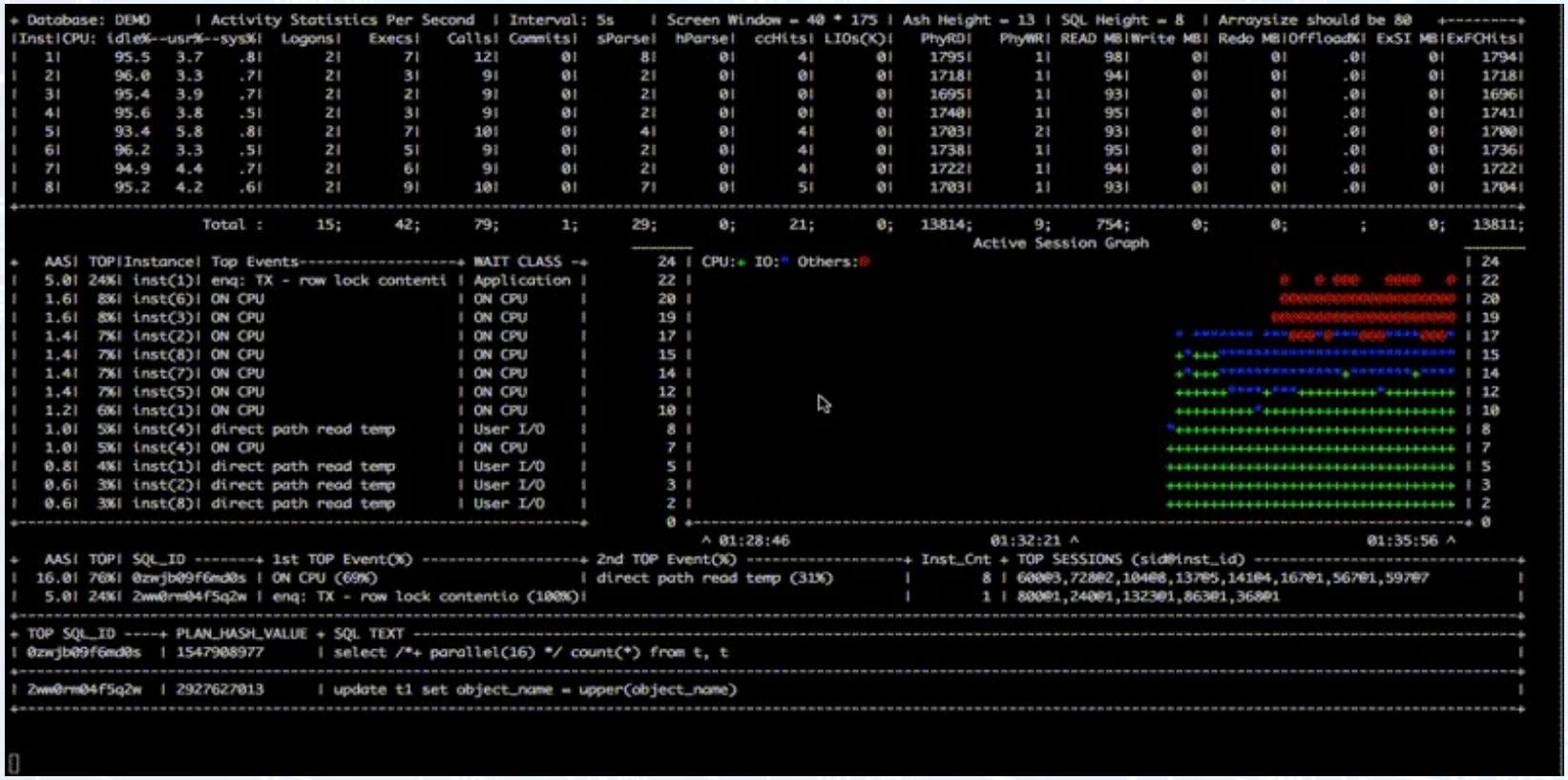
-- Session Snapper v4.24 - by Tanel Poder ( http://blog.tanelpoder.com/snapper ) - Enjoy the Most Advanced Oracle Troubleshooting Script on the Planet! :)


-----------------------------------------------------------------------------------------------------
  ActSes   %Thread | INST | SQL_ID          | SQL_CHILD | EVENT                                | WAIT_CLASS
-----------------------------------------------------------------------------------------------------
    3.79    (379%) |    1 | f2f61y3679ppv   | 2         | ON CPU                               | ON CPU
    1.68    (168%) |    1 | 65ty7zc3cw5cg   | 0         | cell single block read request       | User I/O
    1.00    (100%) |    1 | 8wyfcsarxunt8   | 0         | ON CPU                               | ON CPU
    1.00    (100%) |    1 | auyf8px9ywc6j   | 0         | ON CPU                               | ON CPU
     .95     (95%) |    1 | 6s51jactp16m3   | 0         | ON CPU                               | ON CPU
     .95     (95%) |    1 | 95xtm27vd1f39   | 0         | ON CPU                               | ON CPU
     .89     (89%) |    1 | 5k7t5dmrx47ku   | 0         | ON CPU                               | ON CPU
     .79     (79%) |    1 | b371qcsdvwmg6   | 0         | ON CPU                               | ON CPU
     .74     (74%) |    1 | 71ynst665n0bp   | 0         | ON CPU                               | ON CPU
     .74     (74%) |    1 | ckymf87fs6uy8   | 1         | ON CPU                               | ON CPU

--  End of ASH snap 1, end=2018-05-02 15:04:49, seconds=5, samples_taken=19, AAS=23,6
```

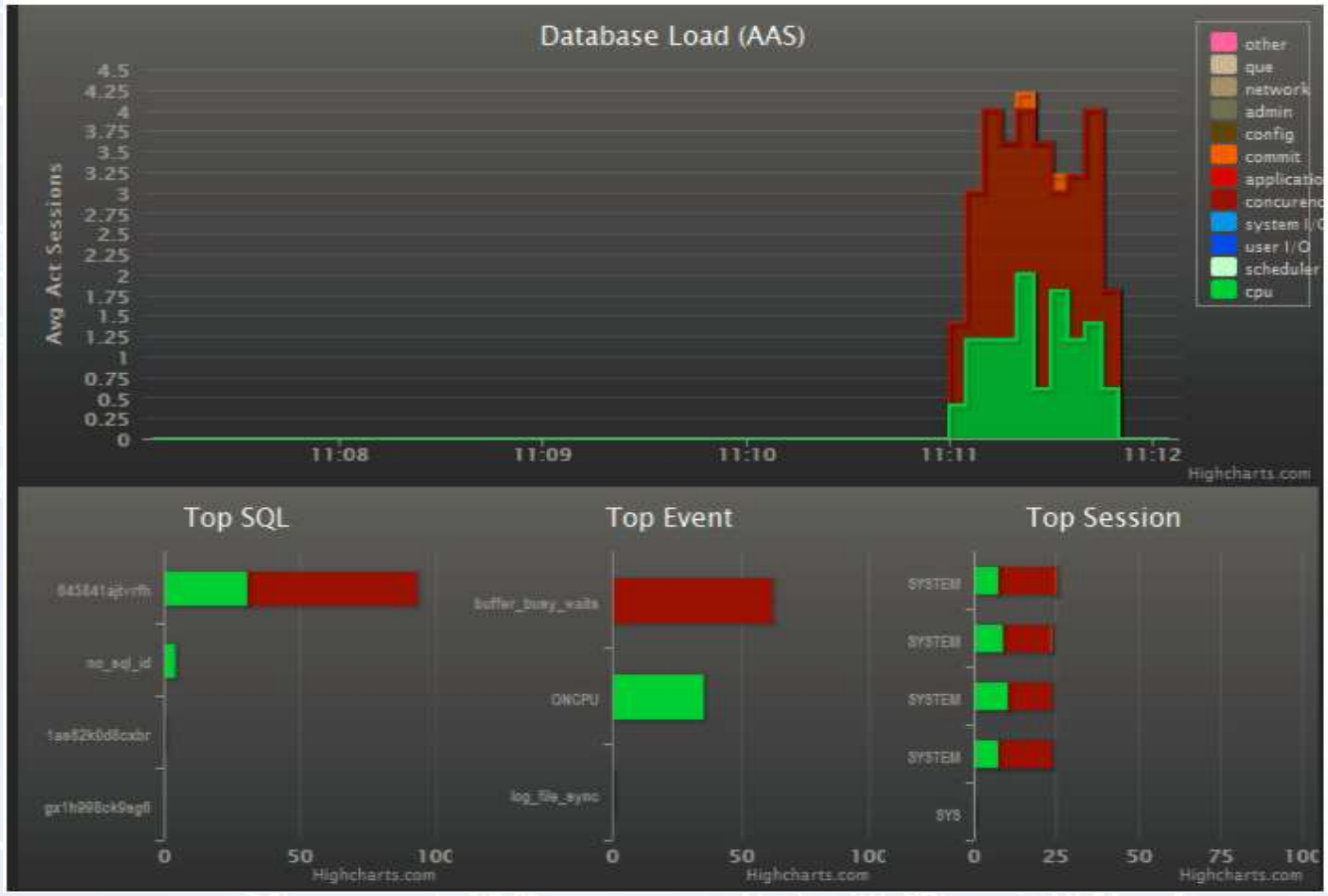
# Real-time analysis – MOATS 2.0

https://github.com/dbsid/moats_rac



# Real-time analysis – Web-ASH

http://datavirtualizer.com/web-ash-w-ash/



# Real-time analysis – TFA(ORATOP)

root@odaora010:/u01/app/12.1.0.2/grid/tfa/odaora010/tfa_home/bin

```
Oracle 12c - Primary PRTCOR 12:32:20 up:  17d,   2 ins, 1.6k sn,   3 us,  17G mt,    5% fra,   0 er,                                3.2% db
ID %CPU LOAD %DCU   AAS  ASC  ASI  ASW  ASP  AST  UST MBPS IOPS IORL LOGR PHYR PHYW  %FR   PGA TEMP UTPS UCPS SSRT DCTR DWTR  %DBT
 1   11    5    0   1.5    0    2    0    0    2 1.6k    7  977   1m 6.1k  914    0    5  145M 3.0M    0 1.1k   1m   14   85  98.3
 2    5    1    0     3    0    0    0    0    0    8    0    3  34u    1    0    0    8  515M 4.0M    0   23 387u   96    3   1.7

EVENT (C)                                                    TOTAL WAITS   TIME(s)  AVG MS  PCT                    WAIT_CLASS
db file sequential read                                        5.499E+08   1681548     5.4   71                    User I/O
DB CPU                                                                      345844           15
direct path read                                                19303674    161955    11.6    7                    User I/O
gc buffer busy acquire                                          29045174    111869     2.8    5                    Cluster
db flash cache single block physical read                       3.764E+08     74903     0.2    3                    User I/O

ID   SID     SPID USERNAME   PROGRAM    SRV  SERVICE   PGA  SQLID/BLOCKER OPN  E/T  STA  STE  WAIT_CLASS  EVENT/*LATCH          W/T
 1  4681    87638 TOTVS12    DBAccess6  DED  SSIM    1.9M  f3pzv03tc3ruc SEL 5.9h  ACT  I/O  User I/O    db file sequential re   2m
```

# Historic Performance

Statspack

- Shipped on 8i (8.1.6) and beyond
- diagnostic tool for instance-wide performance problems
- also supports application tuning activities by identifying high-load SQL statements
- https://oracle-base.com/articles/8i/statspack-8i

# Historic Performance

TUNAS360

- (TUN)ing with (A)ctive (s)essions
- requires no installation and no parameter when executed
- it observes the workload for few minutes and then collects a set of reports on such load
- Run @tunas360.sql

EDB360

- edb360 is a free to use tool to perform an initial assessment of an Oracle database
- pre-check: @sql/awr_ash_pre_check.sql
- Run: @edb360.sql N NULL
  - Parameter 1: Oracle License Pack (required) - [ T | D | N ]
  - Parameter 2: Custom edb360 configuration filename (optional) - 7a-7b

https://carlos-sierra.net/

# SQL Tuning

SQLHC (Oracle)

Run: @sqlhc.sql N 0vy6pt4krb3gm

- Parameter 1: [ T | D | N ] – If tuning/diagnostic/none packages are licensed.
- Parameter 3: SQL_ID

SQLd360 (mauro-pagano.com)

- helps to diagnose SQL statements performing poorly
- input one SQL statement and output a set of diagnostics files
- Installs nothing
- Run: @sqld360.sql 0vy6pt4krb3gm N
  - Parameter 1: SQL_ID
  - Parameter 2: [ T | D | N ] – If tuning/diagnostic/none packages are licensed.

# SQL Tuning

SQLT (Oracle)

- helps to diagnose SQL statements performing poorly
- input one SQL statement and output a set of diagnostics files
- Oracle Diagnostic and/or the Oracle Tuning Packs enhances functionality to the SQLT tool.
  - Installation
    - cd sqlt/install
    - sqlplus / as sysdba @sqcreate

Main methods

- XTRACT - if you know the SQL_ID or the HASH_VALUE of the SQL to be analyzed
- XECUTE - provides more detail than XTRACT, it executes the SQL being analyzed, then it produces a set of diagnostics files
- XTRXEC - This method combines the features of XTRACT and XECUTE. Actually, XTRXEC executes both methods serially
- XTRSBY - Use this method if you need to analyze a SQL executed on a Data Guard or stand-by read-only database.
  You need to know the SQL_ID or the HASH_VALUE of the SQL to be analyzed.
- ~~XPLAIN - This method is based on the EXPLAIN PLAN FOR command~~ use dbms_xplan.display_cursor instead !*
- XPREXT - Use this method if you have used XTRACT and you need a faster execution of SQLT while disabling some SQLT features.
  Script sqlt/run/sqltcommon11.sql shows which features are disabled.
- XPREXC - Use this method if you have used XECUTE and you need a faster execution of SQLT while disabling some SQLT features.
  Script sqlt/run/sqltcommon11.sql shows which features are disabled.

SQLT Main Report: Usage Suggestions (Doc ID 1922234.1)

*https://sqlmaria.com/2017/11/21/does-the-explain-plan-command-really-show-the-execution-plan-that-will-be-used/

# SQL Tuning

Trace Events

10046 – extented trace

i. Level 0: no stats
ii. Level 1: enable standard SQL_TRACE
iii. Leve 4: Level 1 + bind variables
iv. Level 8: Level 1 + wait events
v. Level 12: Level 1 + bind + waits

```
ALTER SESSION SET sql_trace=TRUE;
ALTER SESSION SET EVENTS '10046 trace name context forever, level 8';
EXEC DBMS_SYSTEM.set_sql_trace_in_session(sid=>123, serial#=>1234, sql_trace=>TRUE);
ORADEBUG EVENT 10046 TRACE NAME CONTEXT FOREVER, LEVEL 12;
EXEC DBMS_SUPPORT.start_trace(waits=>TRUE, binds=>FALSE);
```

https://antognini.ch/category/apmtools/tvdxtat/

# SQL Tuning

Trace Events

10053 - examine the internal decisions made by the Cost Based Optimizer

i. ALTER SESSION SET EVENTS '10053 trace name context forever, level 1';
ii. DBMS_SQLDIAG.DUMP_TRACE
iii. ALTER SESSION SET EVENTS 'trace[rdbms.SQL_Optimizer.*][sql:cjk13xfm8ybh7]';

http://www.oaktable.net/contribute/10053-viewer

# OS Tools

1. SYSSTAT
    1. KSAR
2. TFA
    1. OSWATCHER
3. GRAFANA
4. CENTREON
5. ZABBIX
6. PROMETHEUS

# Agradecimentos

1. Coordenação do DBA Brasil 3.0 pelo evento

2. Todos que escolheram prestigiar minha palestra

3. TOTVS


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